

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS ANALYTICS
Practical – II

Roll No.:T052	Name:Divyanjali yadav
Class:TYIT	Batch:01
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Code:-

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset.csv")
print(df.head())
print(df.columns)
```

Output:-

```
...   Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  Lot_Size \
0         1360         2         1         1981    0.599637
1         4272         3         3         2016    4.753014
2         3592         1         2         2016    3.634823
3          966         1         2         1977    2.730667
4         4926         2         1         1993    4.699073

   Garage_Size  Neighborhood_Quality  House_Price
0           0                    5  2.623829e+05
1           1                    6  9.852609e+05
2           0                    9  7.779774e+05
3           1                    8  2.296989e+05
4           0                    8  1.041741e+06
Index(['Square_Footage', 'Num_Bedrooms', 'Num_Bathrooms', 'Year_Built',
      'Lot_Size', 'Garage_Size', 'Neighborhood_Quality', 'House_Price'],
      dtype='str')
```

Q.1 Find the correlation between Year_Built and House_Price.

Code:-

```
correlation = df["Year_Built"].corr(df["House_Price"])
print("Correlation between Year_Built and House_Price:")
print(correlation)
```

Output:-

```
... Correlation between Year_Built and House_Price:
0.051967361931064486
```

Q.2 Create a correlation matrix for all numerical columns.

Code:-

```
correlation_matrix = df.corr(numeric_only=True)
print("Correlation Matrix:")
print(correlation_matrix)
```

Output:-

```
... Correlation between Year_Built and House_Price:
0.051967361931064486
```

Q.3 Perform linear regression using:

X = Year_Built

Y = House_Price

Display the coefficient and intercept.

Code:-

```

X = df[["Year_Built"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
coefficient = model.coef_[0]
intercept = model.intercept_
print("Linear Regression: Year_Built -> House_Price")
print("Coefficient:", coefficient)
print("Intercept:", intercept)
print("\nRegression Equation:")
print(f"House_Price = {coefficient:.4f} * Year_Built + {intercept:.4f}")

```

[16] ✓ 0.0s

Output:-

```

... Linear Regression: Year_Built -> House_Price
Coefficient: 638.6524884727304
Intercept: -649854.0823295026

Regression Equation:
House_Price = 638.6525 * Year_Built + -649854.0823

```

Q.4 Calculate the R2 score for the regression model:

Year_Built - House_Price

Code:-

```

X = df[["Year_Built"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score: Year_Built -> House_Price")
print(r2)

```

[15] ✓ 0.0s

Output:-

```

... R2 Score: Year_Built -> House_Price
0.0027006067060743044

```

Q.5 Calculate the R2 score for:

Square_Footage - House_Price

Code:-

```

X = df[["Square_Footage"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score: Square_Footage -> House_Price")
print(r2)

```

[14] ✓ 0.0s

Output:-

```

... R2 Score: Square_Footage -> House_Price
0.9825992634147099

```

Q.6 Calculate the R2 score for:
Neighborhood_Quality - House_Price
Code:-

```

X = df[["Neighborhood_Quality"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score: Neighborhood_Quality -> House_Price")
print(r2)

```

[13] ✓ 0.0s

Output:-

```

... R2 Score: Neighborhood_Quality -> House_Price
6.037338916142776e-05

```

Q.7 Compare the R2 scores of all three regression models.
Code:-

```

# Model 1: Year_Built -> House_Price
X1 = df[["Year_Built"]]
Y1 = df[["House_Price"]]
model1 = LinearRegression()
model1.fit(X1, Y1)
r2_1 = r2_score(Y1, model1.predict(X1))
# Model 2: Square_Footage -> House_Price
X2 = df[["Square_Footage"]]
Y2 = df[["House_Price"]]
model2 = LinearRegression()
model2.fit(X2, Y2)
r2_2 = r2_score(Y2, model2.predict(X2))
# Model 3: Neighborhood_Quality -> House_Price
X3 = df[["Neighborhood_Quality"]]
Y3 = df[["House_Price"]]
model3 = LinearRegression()
model3.fit(X3, Y3)
r2_3 = r2_score(Y3, model3.predict(X3))
# Compare R2 Scores
r2_scores = pd.DataFrame({
    "Regression Model": [
        "Year_Built -> House_Price",
        "Square_Footage -> House_Price",
        "Neighborhood_Quality -> House_Price"
    ],
    "R2 Score": [
        r2_1,
        r2_2,
        r2_3
    ]
})
print("Comparison of R2 Scores:")
print(r2_scores)
print("\nR2 Scores from highest to lowest:")
print(r2_scores.sort_values(by="R2 Score", ascending=False))

```

[12] ✓ 0.0s

Output:-

```
... Comparison of R2 Scores:

      Regression Model  R2 Score
0      Year_Built -> House_Price  0.002701
1      Square_Footage -> House_Price  0.982599
2 Neighborhood_Quality -> House_Price  0.000060

R2 Scores from highest to lowest:

      Regression Model  R2 Score
1      Square_Footage -> House_Price  0.982599
0      Year_Built -> House_Price  0.002701
2 Neighborhood_Quality -> House_Price  0.000060
```